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HSL

HALF YEARLY EXAMINATION - 2025

12 - Std

CHEMISTRY

Time : 3.00 hrs. Marks : 70

PART - I

I **Note : i) Answer all the questions.** 15 X 1 = 15
ii) Choose the most appropriate answer from the gives four alternatives and write the option code and corresponding answer.

1. Minimum TFM value for Grade - I soaps as per BIS standard is ..... .
   a) 70% b) 60% c) 76% d) 56%
2. Roasting of sulphide ore gives the gas (A). (A) is colourless gas. Aqueous solution of A is acidic. The gas (A) is
   a) CO_2 b) SO_3 c) SO_2 d) H_2S
3. The radius of an atom is 300pm, if it crystallizes in a face centered cubic lattice, the length of the edge of the unit cell is
   a) 488.5 pm b) 848.5 pm c) 884.5 pm d) 484.5 pm
4. Carbolic acid is
   a) phenol b) Picric acid c) Benzoic acid d) Phenyl acetic acid
5. Which of these is not a monomer for a high molecular mass silicone polymer?
   a) Me_3SiCl_3 b) Me_2SiCl_2 c) $PhSiCl_3$ d) Me_3SiCl
6. If 75% of a first order reaction was completed in 60 minutes. 50% of the same reaction under the same conditions would be completed is
   a) 20 minutes b) 30 minutes c) 35 minutes d) 75 minutes
7. Which of the following amines does not undergo acetylation?
   a) t - butylamine b) ethylamine c) diethyl amine d) triethyl amine.
8. Permanganate ion changes to ..... in acidic medium.
   a) MnO_4^{2-} b) MnO_2 c) Mn^{3+} d) Mn^{2+}
9. The catalyst used for hydrogenation of oils to obtain vanaspathi
   a) Fe b) V_2O_5 c) Ni d) Mg
10. Which one of the following reaction is an example of disproportionation reaction?
    a) Aldol condensation b) Cannizaro reaction c) Benzoin condensation d) none of these
11. Which of these is not likely to act as Lewis base?
    a) BF_3 b) PF_3 c) CO d) F^-
12. Glucose and Mannose are epimers at ..... .
    a) C3 carbon c) C4 Carbon c) C1 Carbon d) C2 Carbon
13. The molarity of given orthophosphoric acid solution is 2m. Its normality is
    a) 6N b) 4N c) 2N d) None of these
14. Among the following complexes, which one shows zero crystal field stabilization energy (CFSE) is
    a) $[Mn(H_2O)_6]^{3+}$ b) $[CO(H_2O)_6]^{2+}$ c) $[Fe(H_2O)_6]^{3+}$ d) $[CO(H_2O)_6]^{3+}$
15. The number of electrons that have a charge of 9650 coulombs is
    a) 6.22×10^{23} b) 6.022×10^{24} c) 6.022×10^{22} d) 6.022×10^{-34}

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PART - II

Note : Answer any six questions. Question No. 24 is compulsory. 6 X 2 = 12

16. Give a method for refining Nickel.
17. How will you convert boric acid to boron nitride?
18. What are covalent solids? Give example.
19. Give two examples for zero order reaction.
20. State Faraday's second law.
21. Explain Kolbe's reaction.
22. What are food preservatives?
23. Write the uses of Helium.
24. What is called Zwitter ion? Give an example.

PART - III

Note : Answer any six only. Question No. 33 is compulsory. 6 X 3 = 18

25. What are interstitial compounds? Give example.
26. Differentiate - Double salt and coordination compound.
27. Write note on - Lithium - ion battery.
28. Write the tests to differentiate alcohol and phenol.
29. Explain the vulcanization of Rubber.
30. Write Fisher tropsch synthesis.
31. The rate constant for a first order reaction is $1.54 \times 10^{-3}\ S^{-1}$. Calculate its half life time.
32. Write a note on electro osmosis.
33. Write the following conversions. i) Phenyl methanal $\rightarrow$ Benzoic acid
    ii) Phenyl methanal $\rightarrow$ Benzoin iii) Benzene $\rightarrow$ Benzophenone

PART - IV

Answer all the question. 5 X 5 = 25

34. a) i) What is smelting. Give example.
    ii) Write the Limitations of Ellingham diagram.
    b) i) Explain the preparation of Xenon fluoride compounds.
    ii) How will you prepare chlorine in the Laboratory?
35. a) Compare Lanthanides and actinides. ***(OR)***
    b) Derive an expression for Ostwald's dilution law.
36. a) Describe the nature of bonding in metal carbonyls. ***(OR)***
    b) Explain the preparation of colloids by condensation methods.
37. a) Explain Schottky defects and Frenkel defects. ***(OR)***
    b) Formic acid is powerful reducing agent. - Explain.
38. a) i) Humans cannot use cellulose as food. Why?
    ii) Write the classifications of Lipids. ***(OR)***
    b) An automatic nitro compound (A) on reduction with Sn|HCl gives compound (B) C_6H_7N. Which on treatment with benzoyl chloride in the presence of pyridine to give compound (C). Compound (B) on treatment with CH_3Br to give compound (D). Identify compound A,B,C and D. Write the reactions.

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HALF YERLY EXAM - 2025 (SALEM)

XII - CHEMISTRY - KEY

PART - I

1.) c) 76%
2) c) SO_2
3) b) 848.5 pm
4) a) Phenol
5) d) Me_3SiCl
6) b) 30 minutes
7) d) triethyl amine
8) d) Mn^{2+}
9) c) Ni
10) b) Cannizaro reaction
11) a) BF_3
12) d) C_2 carbon
13) a) 6N
14) c) $[Fe(H_2O)_6]^{3+}$
15) c) 6.022×10^{22}

PART - II

16. $Ni_{(s)} + 4CO_{(g)} \xrightarrow{350K} [Ni(CO)_4]_g \xrightarrow{460K} Ni_{(s)} + 4CO_g$

17. $H_3BO_3 + NH_3 \xrightarrow[800-1200K]{\Delta} BN + 3H_2O$

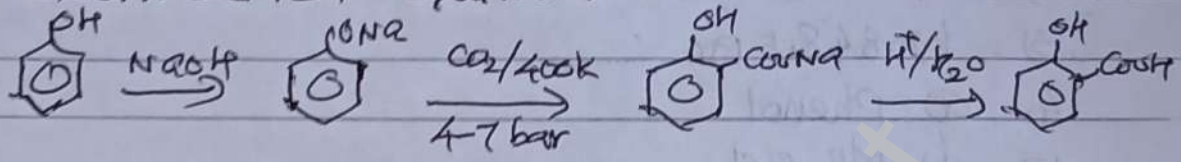
18. The constituents (atoms) are bound together in a three dimensional network entirely by covalent bond
(E.g) Diamond

19. (i) $H_{2(g)} + Cl_{2(g)} \xrightarrow{h\nu} 2HCl_{(g)}$

(ii) $CH_3-\overset{O}{\overset{\|}{C}}-CH_3 + I_2 \xrightarrow{H^+} CH_3-\overset{O}{\overset{\|}{C}}-CH_2I + HI$

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20. The same quantity of charge is passed through the solutions of different electrolytes, the amount of substances liberated at the respective electrodes are directly proportional to their electrochemical equivalent

21. 

22. Capable of inhibiting, retarding or arresting the process of fermentation, acidification or other decomposition of food by growth of microorganism
(Eg) Benzoic acid, sorbic acid

23. * He-oxygen mixture insisted of Air-oxygen mixture to prevent painful dangerous condition called bends.
* Used as electric arc welding of metal
* It has lowest boiling point hence used as cryogenics. * filling air balloons

24. * Molecule having both positive and negative charges
* It is neutral and amphoteric behaviour
These ions are called zwitter ion
(Eg) 

25. PART-III

25. * Compound formed when small atoms like H, B, C or N are trapped in the interstitial holes in a metal Lattice.
(Eg) TiC, Mn_4N

26.

Double salt	Co-ordination compounds
* Contain two simple salt	* formed from Lewis acid and Lewis base
* lose their Identity in solution	* do not completely lose their Identity in Solution
* Mohr's salt	* $K_4[Fe(CN)_6]$

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(27) Lithium ion Battery

* Anode : Porus graphite * cathode : transition metal oxide such as CoO_2

* Electrolyte : Lithium salt in organic solvent

Reaction at anode

$$Li_{(s)} \longrightarrow Li^+_{(aq)} + e^-$$

Reaction at cathode

$$Li^+_{(aq)} + CoO_{2(s)} + e^- \rightarrow LiCoO_{2(s)}$$

overall reaction

$$Li_{(s)} + CoO_{2(s)} \longrightarrow LiCoO_2$$

Uses:

cellular phone, laptops

(28)

	Alcohol	Phenol
* Neutral $FeCl_3$	No colour	purple colour
* $C_6H_5N_2Cl$	No reaction	Red orange dye
* NaOH	No answer	Sodium phenoxide.

(29) Valcanization

* natural rubber $\xrightarrow[100-150^\circ C]{3-5\%\ S}$ cis-1,4 Polyisoprene (C-S-S- bond)

* rubber had become strong and elastic

* rubber $\xrightarrow{3-10\%\ S}$ harder but flexible rubber

(30)

(i) $nCO + (2n+1)H_2 \xrightarrow[\text{Less than 50 atm}]{500-700\ K} C_nH_{2n+2} + nH_2O$

(ii) $nCO + 2nH_2 \xrightarrow{\quad\text{"}\quad} C_nH_{2n} + nH_2O$

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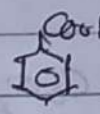
(31) $T_{1/2} = \frac{0.693}{K}$

$= \frac{0.693}{1.54\ s^{-1}} \times 10^3 = \frac{693}{1.54} = 450\ s$

(32) Movement of dispersion medium under the influence of electric potential is called electro osmosis

* Medium moves in a direction opposite to that of the sol particle

(33)

(i) $C_6H_5CHO \xrightarrow[(O)]{Cl_2/KMnO_4}$ 

(ii) $C_6H_5-\underset{H}{\overset{O}{\overset{\|}{C}}} + H-\overset{O}{\overset{\|}{C}}-C_6H_5 \xrightarrow{Alcohol} C_6H_5-CH(OH)-\underset{O}{\overset{\|}{C}}-C_6H_5$

(iii) $C_6H_5-\overset{O}{\overset{\|}{C}}-Cl + C_6H_6 \xrightarrow[FeCl_3]{Anhydrous} C_6H_5-\overset{O}{\overset{\|}{C}}-C_6H_5 + HCl$

PART-IV

(34) (a) (i) Smelting

* Metal oxide is converted into crude metal by using flux and reducing agent is called smelting

(Eg) 1

$Fe_2O_{3(s)} + 3CO_{(g)} \longrightarrow 2Fe(s) + 3CO_2(g)\uparrow$

$CaO_{(s)} + SiO_{2(s)} \longrightarrow CaSiO_{3(s)}$

(Flux) (Gangue) (Slag)

(Eg) 2.

Extraction of copper from copper pyrites

(b) (ii) Limitation of ellingham diagram

* It gives information about the thermodynamic feasibity only

* It does not tell anything about the rate of the reaction

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34. (b) (i) (1) $Xe + F_2 \xrightarrow[400^\circ C]{Ni} XeF_2$

(ii) (2) $Xe + 2F_2 \xrightarrow[400^\circ C]{Ni/\text{acetone}} XeF_4$

(3) $Xe + 3F_2 \xrightarrow[400^\circ C]{Ni/200\,atm} XeF_6$

(iii) $4NaCl + MnO_2 + 4H_2SO_4 \rightarrow Cl_2 + MnCl_2 + 2H_2O + 4NaHSO_4$

(35) (a)

Lanthanoids	Actinoids
(i) e^- Enters in 4f orbital	(i) In 5f orbital
(ii) B.Energy high	(ii) B.Energy is low
(iii) Less tendency to form Complex	(iii) greater tendency to form Complex
(iv) Most of ions Colourless	(iv) Most ions are coloured
(v) do not form Oxocation	(v) form Oxocations.

(b)

* $CH_3COOH \rightleftharpoons CH_3COO^- + H^+$

* $K_a = \dfrac{[CH_3COO^-][H^+]}{[CH_3COOH]}$ $\quad [CH_3COOH] = c(1-\alpha)$, $[H^+] = c\alpha$, $[CH_3COO^-] = c\alpha$

* $K_a = \dfrac{c\alpha \cdot c\alpha}{c(1-\alpha)}$

$$\boxed{K_a = \frac{\alpha^2 c}{1-\alpha}} \Rightarrow (1-\alpha) = 1$$

* $K_a = \alpha^2 c$; $\alpha^2 = \dfrac{K_a}{c} \Rightarrow \alpha = \sqrt{K_a/c}$

* $[H^+] = \left(\sqrt{\dfrac{K_a}{c}}\right) c = \sqrt{\dfrac{K_a c^2}{c}}$

$[H^+] = \sqrt{K_a c}$ similar to weak base

$[OH^-] = \sqrt{K_b c}$

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36. (a) Bonding in metal carbonyl

* $M \xleftarrow{\sigma} CO$
(Vacant d-orbital) (Filled orbital)

* σ bond formation increase the electron density in metal 'd' orbital

* Increase the electron density, a filled metal d-orbital interacts with the empty π^* orbital on the carbonyl ligand and transferred back to ligand
This is called π back bonding

* $M \xrightarrow{\pi} CO$

(b) Condensation Method

(i) oxidation

$2H_2Se + O_2 \rightarrow 2H_2O + 2Se_{(sol)}$

(ii) Reduction

$2AuCl_3 + 3HCHO + 3H_2O \rightarrow 2Au_{(s)} + 6HCl + 3HCOOH$

(iii) Hydrolysis

$FeCl_3 + 3H_2O \rightarrow Fe(OH)_3 + 3HCl$

(iv) Double decomposition

$As_2O_3 + 3H_2S \rightarrow As_2S_3 + 3H_2O$

(v) Decomposition

$S_2O_3^{2-} + 2H^+ \rightarrow \underset{(s)}{S} + H_2O + SO_2$

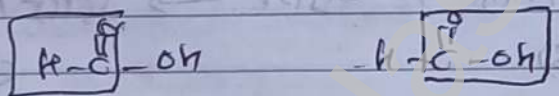
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(37) (a) (i) Schottky defect

* Missing equal no of C^+ and A^-
* Does not change stoichiometry
* similar size C^+ and A^- (Almost)
* Lowers its density (Eg) NaCl

(ii) Frenkal defect

* Arises to dislocation of ions from its (crystal) Lattice
* occupies of ion an interstial position
* differ size C^+ and A^-
* does not effect density
* Eg AgBr

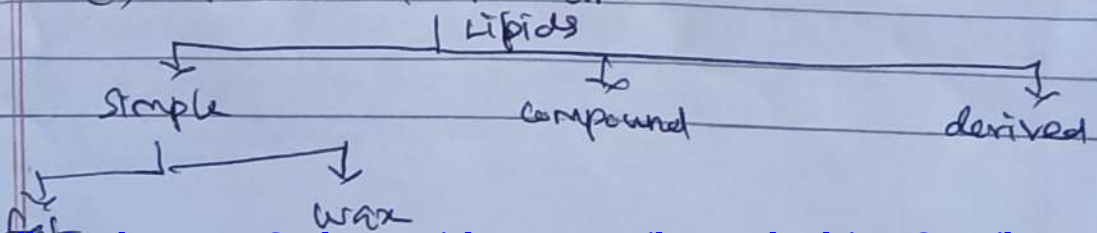
(b) * 

* $H{-}CHO^- + 2Ag^+ + 3OH^- \rightarrow 2Ag + CO_3^{2-} + 2H_2O$
(Tollen's reagent) (silver mirror)
* $H{-}CHO^- + 2Cu^{2+} + 5OH^- \rightarrow Cu_2O + CO_3^{2-} + 3H_2O$
Fehling solution (Blue) (Red)

(38) (a)

(i) our digestive systems do not contain the necessary enzymes (glycosidase (or) cellulases)

(ii) Based on structure



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* Based on esters group

- Phospholipids (Phospo ester linkage)
- glycolipids (Sugar) Molecule
- Lipo proteins (protein)

(b)

(i) $C_6H_5NO_2 \xrightarrow{Sn/HCl} C_6H_5NH_2$

(A) (B)

(ii) $C_6H_5NH_2 + Cl-\overset{O}{\overset{\|}{C}}-C_6H_5 \xrightarrow[Py]{Neech} C_6H_5NH-\overset{O}{\overset{\|}{C}}-C_6H_5$

(B) (C)

(iii) $C_6H_5NH_2 + CH_3Br \longrightarrow C_6H_5NHCH_3$

(B) (D)

A	$C_6H_5NO_2$	Nitro benzene
B	$C_6H_5NH_2$	Aniline
C	$C_6H_5NHCOC_6H_5$	N-Phenyl benzamide
D	$C_6H_5NHCH_3$	N-Methyl Aniline

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